

Auditable Legal-Action Learning for Partially Observable Trading Card Games: Causal Representation Repair and Receipt-Backed Evaluation

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Abstract—Trading card game agents act through variable, multi-stage legal menus under partial observability. This makes representation correctness inseparable from policy quality: two options with different public rule semantics may be aliased by a transient option index, while a richer representation can accidentally leak an opponent’s private state. We describe a direct-policy Pokémon TCG controller that keeps the pinned simulator as the sole legality and transition authority, restricts inference to the acting player’s information set, and adds a zero-bypass public-rule semantic residual to a preserved causal policy. A content-addressed 20-day corpus and collision census drive branch activation; unsupported metadata, auxiliary, and checklist routes remain exact-zero. The census contains 18,412,973 option records and identifies 127,641 collision groups in the legacy projection. The repaired key leaves zero unresolved actionable semantic collisions, with 99 residual duplicate groups classified as harmless permutations. We further specify full-model bootstrap, direct-policy self-play, a trainer-only public Prize-plan canary, matched evaluation, and immutable receipt lineage. A 900-game iteration-2 holdout is reported with its non-promotion pipeline disposition, and an iteration-3 holdout waiver is reported as zero measured games. The paper’s central claim is therefore representation and audibility, not leaderboard superiority or causal gameplay improvement.

Index Terms—reinforcement learning, partial observability, legal-action representation, trading card games, reproducibility, semantic collisions, policy audit

I. INTRODUCTION

A digital trading card game exposes a compact interface—a state and a list of legal menu choices—but the underlying decision is neither a fixed discrete action nor a fully observed Markov state. Search, recovery, target selection, ordered multi-select, and optional stopping can create several prompts for one conceptual action. The meaning of an option depends on its source, target, count, effect, and the public state in which the engine produced it. Meanwhile, each player knows private facts that the other does not. These properties make the problem naturally related to partially observable sequential decision making [1], [2], but they also create software-specific failure modes that a generic POMDP description does not capture.

The first failure mode is *legal-option aliasing*. A policy may receive an option index or a lossy payload in which

distinct public choices collapse to the same representation. Increasing model capacity cannot recover a distinction that is absent from the input. The second is *information leakage*. Fields that are useful as offline targets—future transitions, terminal outcomes, opponent private cards, or unrevealed Prize identities—must not become deployment features. The third is *claim leakage*: training accuracy, public leaderboard score, shadow-planner diagnostics, and formal held-out win rate are measurements of different objects. Combining them into one apparent performance result produces a paper that is stronger than its evidence.

This work describes the finalized design of an isolated Alakazam rule-derivative controller governed by a versioned, typed contract [3]. The controller is direct policy at deployment. The official pinned simulator enumerates legal options and executes the selected current index; no learned component can make an illegal option legal. A causal state/history encoder and the legacy card representation are preserved. Beside them, a separately versioned residual represents exact public rule semantics: selection context, numeric value, count range, remaining resource budget, effect, and stable source/target/area/slot bindings. New branches begin at exact zero and become trainable only if a full corpus census supports them. This turns representation migration into an empirical, reversible change rather than an architectural assertion.

The resulting paper is organized around evidence boundaries. Its contributions are:

- an acting-player information-set contract that separates runtime fields, target-only fields, and forbidden hidden truth;
- a factorized legal-action representation in which semantic repair is additive, zero-bypassable, and subordinate to simulator legality;
- a corpus-driven activation rule that accepts only the public-rule projection and keeps unsupported metadata, repaired-head, and checklist gates inert;
- a checksum-bound training lifecycle covering a 25-epoch full-model bootstrap, direct-policy self-play, expert refresh, and a trainer-only H3 Prize-plan canary;
- paired evaluation and claim-status conventions that pre-

serve failed, waived, historical, and not-yet-measured results instead of relabeling them.

The strongest measured technical result is a collision repair over 18.4 million option records. Gameplay observations are included because a systems paper should expose its current evidence, but their scope is narrow. A Kaggle score is not treated as a controlled ablation; a formal holdout whose broader gate failed is not presented as promotion; and a waived holdout with zero games is not a pass. This conservative separation follows reproducibility practice in which the experimental unit, data lineage, and reporting decision are explicit [4].

II. PROBLEM FORMULATION AND THREAT MODEL

A. Information set and legal action set

Let I_t be the complete information legally available to the acting seat at decision time t . It includes the acting player’s own hand and other legitimately private state, the public board and zones, public action history, legally revealed card information, and the current legal-option list. It does not include the opponent’s hidden hand identities, either deck order, or unrevealed Prize identities. A deterministic observation builder produces $x_t = f(I_t)$ and a causal history $h_t = (x_{\leq t}, a_{< t})$.

The pinned simulator returns a variable set

$$\mathcal{A}_t = \mathcal{A}(I_t) = \{o_{t,1}, \dots, o_{t,n_t}\}. \quad (1)$$

The policy chooses only among these current options. Some conceptual actions span several engine prompts, so a complete action is a finite sequence $a_t = (o_{t,i_1}, \dots, o_{t,i_K}, \text{stop})$ in which the simulator recomputes legality after each stage. The policy never caches an old option index as a future command. This distinction matters because an index is an address inside one transient list, not an action identity.

We use the terminal return $z \in \{-1, 0, +1\}$ for loss, draw, and win. Auxiliary targets may describe public resources, Prize potential, or short-horizon outcomes, but they are masked when the required causal interval is incomplete or ambiguous. Missing supervision is not encoded as a numerical zero.

B. Threat model

We consider four classes of failure.

Representation collision. Two publicly distinguishable options with different simulator semantics share one feature key. Harmless permutations are not actionable collisions; the criterion is a semantic difference, not byte inequality.

Privileged-input dependence. A pre-transition choice changes when only hidden opponent cards, deck order, unrevealed Prizes, or future events are changed. Metamorphic pairs must leave sanitized observations, option features, and policy logits identical.

Authority inversion. A learned rule or legality predictor adds, removes, or rewrites simulator options. Such a model may be a diagnostic head, but it is never an execution authority.

Evidence conflation. An integration receipt, training diagnostic, public score, descriptive replay slice, or shadow-model metric is reported as a formal policy result. We control this with visible claim labels—**Measured**, **Receipt-verified**,

Configured, **Historical**, and **Not measured**—and with receipt-bound tables whose disposition accompanies the number.

C. Scope

The active derivative uses a greedy, factorized policy. RTP, MCTS, online search, and the recursive PokerRLM roadmap are excluded from the packaged action path. A separately trained shadow planner may supply research diagnostics, but its presence does not imply action authority. Likewise, the Prize-plan v2 sidecar is a trainer-only advantage provider. It is absent from the policy state dictionary, runtime Fusion, selector, serving bundle, and Kaggle package.

III. SYSTEM DESIGN

A. Simulator-owned legality

The environment boundary is intentionally asymmetric. The simulator owns exact rules, random outcomes, legal menus, transitions, and terminal conditions. The controller owns representation and ranking. At every factorized stage the wrapper requests the current legal list, encodes every option, selects one current index, and submits that index. If model load or inference fails, a seeded fallback still chooses from the supplied list. This cannot guarantee a strategically good action, but it preserves the basic legality invariant

$$a_{t,k} \in \mathcal{A}(I_{t,k}) \quad \text{for every executed stage } k. \quad (2)$$

B. Causal state and option encoding

The base model uses a Transformer-style causal representation [5]. Spatial board information, acting-seat private state, public flags and counts, and realized action history are encoded without future tokens. Each legal option receives structured identity, category, source, destination, target, count, effect, and continuation context. A factorized decoder compares the current legal set in parallel, then emits one option at a time. For multi-select prompts, already selected options and the remaining budget are part of the next stage; an explicit stop becomes selectable only when the engine exposes it.

The score for current option i can be written schematically as

$$\ell_i = \ell_i^{\text{base}} + g_{\text{rule}} r_i + \sum_{j \in \mathcal{H}} m_{j,i} \rho_{j,i} \Delta_{j,i}, \quad (3)$$

where r_i is the rule-semantic residual, g_{rule} is its gated activation, and each auxiliary route has an availability mask $m_{j,i}$, bounded reliability $\rho_{j,i}$, and option-conditioned contribution $\Delta_{j,i}$. Equation (3) describes the architectural contract, not a claim that every physical route is active. The census accepted only the public-rule projection; unsupported branches keep exact-zero gates.

C. Public-rule semantic repair

The legacy card representation is preserved bit-for-bit. The repair is a parallel residual whose schema includes:

- exact NUMBER values rather than a coarse bucket or text hash;
- selection context, effect, context card, and option count;

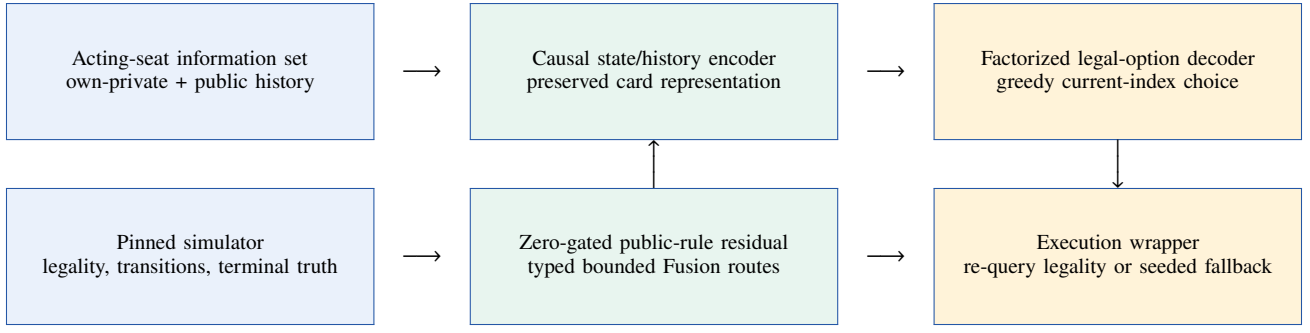


Fig. 1. Final direct-policy system boundary. The simulator is authoritative; the policy ranks only the current legal options. Prize-plan critics, human guide, RTP, and recursive planning do not provide runtime action logits.

Runtime-eligible	Target-only or forbidden
Own hand and legally known private state	Opponent hidden hand identities
Public board, zones, flags, counts	Either deck's future order
Public history and current legal options	Unrevealed Prize identities
Pinned public catalog metadata	Future transitions and terminal result

Fig. 2. Information-flow boundary. Offline labels remain separate typed objects and cannot enter the deployment observation.

- minimum, maximum, selected, and remaining selection counts;
- remaining damage and energy budgets when publicly defined;
- stable source, target, area, slot, tool, and energy bindings;
- public current/maximum HP, evolution stack, appearance-this-turn, bench maximum, typed energy units, and once-per-turn flags when retained and proven;
- pinned public catalog metadata through a separately gated adapter.

Global serials and incidental menu order are not semantic identities. Stable within-observation entity bindings are preferred; ordinal position is a last-resort fallback only when the simulator distinguishes otherwise identical payloads. Text hashes may remain in the legacy card path but never serve as exact mechanics authority for the new residual. The new path is initialized to zero, and a layer-off test requires bit-identical base logits. This gives the migration an executable rollback rather than a prose promise.

D. Corpus-driven branch adjudication

Implementation is not evidence of usefulness. Every proposed branch is first physical, default-zero, and inert. The frozen schema is then applied to the complete eligible corpus, and a census measures public semantic collisions, selected-action involvement, deck and matchup distribution, transition witnesses, and harmless equivalences. Only a branch supported by the census may enter training or action scoring.

For the current snapshot, adjudication accepted `public_rule_semantic_projection`. The public metadata residual, repaired auxiliary heads, and all eight checklist-provenance gates remained unsupported or unproven and

therefore exact-zero. This distinction prevents a paper from describing a head inventory as realized decision influence.

E. Checklist provenance

The schema records eight public questions at every actor-visible decision or forced selection: current knockout hand threshold; cards safely spendable above that threshold; replacement Alakazam line; unavoidable draws before the next attack; bench Prize exposure; immediate disruption outcome; a robust line under unknown Prize identity; and whether the game ends before a forced draw. Each channel records factual evidence, availability, an unavailable reason, per-option signed evidence when defined, the gate, and final residual. Unknown or hidden evidence yields zero. The current corpus did not authorize the checklist gate branch, so these records support trace and future calibration, not a present action-influence claim.

F. Typed auxiliary separation

The r274-compatible policy structure contains existing policy, value, strategic-head, Fusion, own-deck, and matchup paths. Revision 10 mapped immutable r195-compatible weights into that structure and optimized every architecture-present non-combo model weight. Combo tensors were retained only for checkpoint and pack ABI compatibility, with zero loss, no optimizer membership, and a disabled runtime route. The preserved Card2Vec branch is a specific non-regression boundary: the semantic residual may accompany it but cannot replace, re-embed, or rewire it.

Matchup routing is also causal. A route must resolve from supported public identity; unknown, ambiguous, retired, or unsupported identities take the base-policy bypass. Model materialization alone is not route selection, and route selection alone is not evidence of improved outcomes.

IV. DATA AND TRAINING LINEAGE

A. Eligibility and split isolation

Revision 9 selected the most recent 20 contiguous UTC days, 23 July through 11 August 2026. Raw episodes are scanned, but derivative rows are restricted at the acting-seat level. A decision is eligible if and only if that seat's exact recorded setup deck contains literal card ID 743. This is a membership test, not an archetype classifier: a tech copy qualifies, and a

TABLE I
COMPONENT STATE AT THE EVIDENCE SNAPSHOT. PHYSICAL PRESENCE IS NOT EQUATED WITH TRAINING, RUNTIME ACTIVATION, OR MEASURED BENEFIT.

Component	Physical	Derivative training	Runtime action path	Boundary
Causal backbone, policy/value, existing Fusion	Yes	Eligible under rev. 10	Yes	Acting-player-visible inputs; current legal set only
Preserved Card2Vec path	Yes	Preserved	Yes	Bit identity and layer-off parity; not replaced by rule residual
Public-rule semantic projection	Yes	Accepted	Yes	Zero-initialized additive residual; simulator remains authoritative
Public catalog metadata residual	Yes	No	No	Census branch unsupported; exact-zero/inert
Repaired auxiliary heads	Yes	No	No	Census branch unsupported; exact-zero/inert
Eight checklist-provenance gates	Schema/trace	No	No	Evidence emitted; gate branch unsupported and exact-zero
Combo-state tensors	ABI bytes	No	No	Zero loss; optimizer-excluded; route disabled
Prize-plan v2 sidecar	Separate	Three-epoch sidecar	No	Trainer-only H3 advantage; no policy gradient into sidecar
RTP / recursive planner	Separate shadow/roadmap	Shadow only	No	No package, selector, Fusion, or serving authority

deck label cannot substitute for the card record. If only one seat qualifies, the other seat’s decisions are excluded. Each included row retains a digest of the complete acting-seat setup-deck multiset so list variants can be stratified without becoming hidden runtime inputs.

The sealed corpus contains 24,335 games, 2,351,208 decision occurrences, 18,412,973 option records, and 68 acting-deck variants. Whole UTC days are assigned 14/3/3 to train, validation, and evaluation. The source manifests report zero cross-split overlap by source archive, day, game, and group. The evaluation split is unavailable for fitting, support thresholds, scaling, or tuning. Missing or malformed acting-deck evidence causes explicit exclusion, not guessed eligibility.

Targets come from recorded selected actions and causally aligned public trajectories. The system does not fabricate labels for unchosen actions. Counterfactual simulator data, if later required, must be a separately identified corpus with public-information restoration seeds and its own provenance. Incomplete, ambiguous, nonmonotone, or terminal-censored intervals are masked rather than assigned zero.

B. Full-model bootstrap

The candidate structure maps exact-name/shape-compatible r195 tensors into the r274 head architecture plus bug-fixed derivative paths. All twenty day archives are freshly packed against this structure; an older pre-fix tensor pack cannot substitute. Training runs 25 genuine full-corpus epochs, and each epoch visits every eligible decision occurrence once before the next epoch. Semantic-pattern deduplication may support analysis but cannot replace optimizer rows.

All architecture-present non-combo weights are optimizer-eligible under the revision-10 override. Combo-state bytes remain physical only to preserve an existing checkpoint/package ABI: their loss is zero and their training and runtime routes are disabled. A prior short, sidecar-only run is preserved as invalid historical evidence and cannot seed self-play, routing, or promotion. This negative provenance is important because

a valid checkpoint is defined by its data and optimizer scope, not just by a finite loss.

C. Direct-policy self-play schedule

After validation and one checksum-bound initial package, the derivative enters 20 direct-policy iterations. Each iteration collects exactly 8,196 source games: 1,024 self-play and 7,172 against a checksum-pinned public-package mix. Seats are balanced. At iterations 5, 10, 15, and 20, the loop seals collection and checkpoint state, then runs five full expert-refresh epochs on the same recent-20-day pack with the same non-combo optimizer scope. Kaggle packages are queued only at iterations 10 and 20. There is no iteration 21.

Every shard, pack, checkpoint, optimizer state, opponent package, seed, seat, and boundary action is content-addressed. A remote worker may produce games, but the Inzi Blackwell host remains the sole learner and optimizer. Duplicate suppression and unique lease/game identities prevent one returned game from appearing twice. A host fault reduces eligible capacity; it does not silently change the checkpoint, package, or exact retained count.

D. Trainer-only Prize-plan canary

The Prize-plan v2 sidecar trains horizons $h \in \{1, 3, 6, 12\}$ on complete, recorded, chosen action programs. Public remaining-Prize counts define a train-split-only monotone potential

$$\Phi(p_{\text{ours}}, p_{\text{opp}}) = 2P_{\text{iso}}(\text{win} \mid p_{\text{ours}}, p_{\text{opp}}) - 1. \quad (4)$$

The potential cannot decrease when our remaining count falls and cannot increase when the opponent’s remaining count falls. A receipt-bound fixed γ produces segment reward

$$r_t^P = \gamma\Phi(s_{t+1}) - \Phi(s_t), \quad (5)$$

a potential-based shaping form related to policy-invariant reward transformations [6]. The current actor canary uses only H3:

$$A_{H3}(s, a) = z - V_{\text{existing}}(s) + 0.025 m_3 c_3 \left(Q_{\text{plan}}^3(s, a) - V_{\text{plan}}^3(s) \right). \quad (6)$$

Here $m_3 = 1$ only when all three complete same-seat segments are causal, unambiguous, and not terminal-censored. The scalar $c_3 \in [0, 1]$ is a frozen, receipt-bound public confidence/action-support value fit on training data only. The one complete-action advantage is broadcast identically across its selected factorized stages. Sidecar predictions are precomputed and frozen for an actor unit; no actor gradient reaches the sidecar.

The legacy branch is exactly $z - V_{\text{existing}}(s)$. Any identity, alignment, support, calibration, finiteness, effective-sample-size, clipping, or noninterference failure restores that branch at a clean boundary without rewriting the policy checkpoint. H1, H6, and H12 remain diagnostic under H3 mode with actor coefficient zero; simultaneous horizon terms are forbidden.

V. EVALUATION METHODOLOGY

A. Representation and engine tests

Evaluation begins below gameplay. Layer-off tests require bit-identical base logits, legal choices, and Card2Vec outputs. The collision census classifies every legacy duplicate by public semantic delta, selected-action involvement, and available successor witness. Targeted engine cases cover full-bench attack requirements, deck-zero or opponent-hand-zero conditions, automatic effect ordering, simultaneous-knockout prompt order, ordinary/ex/Mega-ex Prize counts, and visible Prize modifiers. A legality predictor may flag a discrepancy but cannot alter the engine list.

Information-set metamorphic tests hold the acting player's sanitized state, history, and legal options fixed while changing privileged opponent hands, deck order, or unrevealed Prizes. Observation hashes, option features, base logits, candidate logits, and any public-belief proposal must remain identical. The realized future may change after action execution; the pre-transition decision may not depend on that hidden truth.

B. Paired policy comparisons

The smallest reported gameplay comparison is BO250: 125 seed-matched pairs, with candidate and reference swapping seats inside each pair. BO250 is a diagnostic, not a conclusive promotion gate. Reports include wins, draws, losses, paired intervals, action disagreement, severe rule-error rate, residual coverage, and which bounded contribution changed each action.

Formal evaluation uses a fixed opponent roster and 50 games per opponent deck, split 25/25 by candidate seat. With the current 18-deck roster this yields 900 games. We report skill-weighted rate, its receipt-defined lower bound, minimum opponent rate, tier strata, and every pipeline criterion. A statistically favorable aggregate does not erase a failed rating simulation,

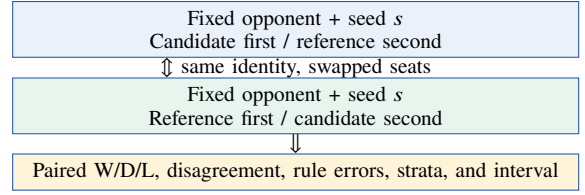


Fig. 3. Seat-swapped matched pair. BO250 contains 125 such pairs; formal and Prize-plan protocols use larger fixed schedules.

historical non-regression check, or identity audit. Wilson-style binomial intervals are useful descriptive summaries [7], but the receipt-defined gate remains the operational criterion.

The Prize-plan study has a separate preregistered structure. Eligible legacy, H1, H3, and later H6 arms each require 8,192 seed-matched, seat-swapped games when capacity permits. Required outputs include W/D/L, Prize conversion, matchup and public-Prize strata, late closeout, deck-out, action entropy, calibration, coverage, AWR effective sample size, clipping, and action disagreement. No such completed multi-arm gameplay receipt exists at the paper snapshot.

C. Evidence tiers

We distinguish five tiers. **Measured** is a completed numerical artifact with its experimental unit. **Receipt-verified** is an integration, identity, or training receipt that does not itself prove game strength. **Configured** is a future protocol or active design. **Historical** is context from a different checkpoint, deck, or uncontrolled public environment. **Not measured** means no result is available, including a waived test. These labels remain adjacent to each table row so a detached number cannot silently acquire a stronger interpretation.

VI. RESULTS AND EVIDENCE STATUS

A. Representation census

The 20-day census scanned 18,412,973 option records over 2,351,208 decisions. The legacy public-semantic key produced 127,641 collision groups containing 255,398 records, or approximately 1.39% of all options. Selected actions appeared in 129,503 of the collision records. Six semantic-delta classes were observed; five had targeted transition witnesses and one was supported by a legality-only witness. After adding the structured projection, the census found 99 raw duplicate groups. Every one was permutation-equivalent, leaving zero unresolved actionable repaired-key collision groups. This result supports the representation repair. It does not, by itself, estimate a win-rate delta.

B. Bootstrap diagnostics

The full bootstrap completed 25 epochs over 2,351,208 decision occurrences and 18,412,973 option occurrences per epoch. This corresponds to approximately 57.88 million decision-occurrence visits in 5,757.14 seconds. The final semantic loss was 1.7141 and semantic accuracy was 42.27%. The sealed checkpoint begins with digest 8b59af9a. These numbers demonstrate the intended optimizer scope and completed workload; semantic accuracy is not gameplay accuracy.

C. Offline critic evidence

The Prize-plan v2 sidecar trained for three epochs and 30,552 optimizer steps. Validation contained 380,499 complete actions, with 358,344 H3-eligible actions (94.18% coverage). H3 action-value mean squared error was 0.0040745 versus 0.0046184 for the empirical-mean baseline, an 11.78% reduction; the state-value error was 0.0045797 versus 0.0046184, a 0.84% reduction. On 2,146,670 exact policy decisions, the receipt recorded legacy and H3 effective-sample-size fractions of 0.251337 and 0.251758, with clip fractions 0.076078 and 0.075819. The non-regression check passed. These are held-out prediction and actor-weight stability results. The sidecar remained outside runtime policy inference.

A separate eight-output action critic also trained for three epochs. All eight held-out outputs beat zero and empirical-mean baselines. For context, Q_{win} recorded Brier score 0.24728 versus 0.24979 and expected calibration error 0.03406; V_{win} recorded 0.24921 and 0.02964. This critic is shadow-only and supplies no serving-policy efficacy claim.

D. Gameplay observations and disposition

The checksum-bound NO-RTP package for submission 55487412 completed with a public score of 600.0. Search and MCTS were disabled. Because there is no seed-matched baseline under the same platform conditions, this is a contextual leaderboard observation rather than causal evidence for the semantic adapter.

Iteration 2 completed a receipt-audited 900-game formal holdout against 18 opponents, 50 games per opponent and a 25/25 candidate-seat split. It recorded a 76.83% skill-weighted win rate, 74.39% reported lower bound, 54.0% minimum opponent win rate, and 76.12% S-tier mean. The broader pipeline gate failed because of Kaggle-rating simulation and historical-nonregression criteria. Accordingly, the checkpoint was carried as a continuous learner rather than reported as a promotion pass.

Revision 27 waived only iteration 3's formal holdout. The candidate and promotion comparison were preserved, but 900 planned games were skipped and zero formal-holdout games were measured. Its promotion comparison was rejected because the confidence lower bound did not clear the threshold. At the snapshot, iteration-4 collection was sealed but no commit or evaluation result existed; a provider-initialization `NameError` occurred before H3 materialization or optimization and was handled through a receipt-bound recovery path. We report this as operational status, not a policy result.

VII. DECK STRATEGY AND INTERPRETABILITY

Alakazam's Powerful Hand provides an unusually interpretable local objective: two damage counters are placed for each card in hand. For a visible target with remaining HP H , the basic hand threshold is

$$h_{\text{KO}} = \lceil H/20 \rceil, \quad (7)$$

modified only by visible, applicable effects. Public deck guides provide useful context [8], [9], but the project guide is training

curriculum metadata, not a serving action source. It cannot read hidden facts, provide policy cross-entropy, or override the learned option score.

The strategy is resource-constrained card-count control, not unconstrained draw. A stable board maintains a current attacker route, a replacement Abra or evolved line, a Dunsparce draw/recycle route when it solves a defined problem, and a flexible bench slot. Deterministic search normally precedes broad draw when the target is already known. Rare Candy is preserved when evolving through Kadabra keeps multiple routes live. Boss's Orders is spent for a prize-changing or tempo-winning target; Enhanced Hammer is used when a visible Special Energy actually prevents damage or changes the prize race. Near deck-out, mandatory turn draw and every unavoidable attachment, evolution, or ability draw are counted before committing the line.

The 14 August matchup supplement illustrates how semantic features connect to human reasoning without introducing hidden inference. A visible 330-HP Mega Lopunny ex requires 17 hand cards for a full-health Powerful Hand knockout and gives up three Prizes. A visibly attached Mist Energy prevents the attack effect, so Enhanced Hammer or a gust-around line is valuable. The controller may use visible HP, card class, attachment, and legal options. It may not infer a hidden Mist Energy or healing card merely because the opponent revealed Buneary.

Interpretability follows the same causal boundary. For each current option the trace can expose the base logit, structured rule residual, masks, bounded route reliabilities, and final score. Leave-one-route-out logit changes measure decision sensitivity, not counterfactual win effect. A route should be reweighted only after paired fixed-seed intervention. Likewise, checklist records explain what evidence was available, but their exact-zero gates at this snapshot mean they did not change the deployed action.

VIII. REPRODUCIBILITY, ETHICS, AND LIMITATIONS

The reproducibility unit is a digest-bound claim rather than a mutable filename. The minimum identity set contains the typed goal contract, corpus and split manifests, feature/target/checklist schemas, simulator and catalog build, checkpoint ancestry, optimizer scope, deck multiset, runtime tree, opponent packages, seed/seat schedule, and evaluation receipt. Package manifests bind every archive member. Superseded checkpoints and failed attempts remain immutable evidence and cannot be renamed into an active result. This approach is consistent with the broader need for explicit experimental artifacts and reporting decisions in machine learning [4].

The official Kaggle simulation environment [10] is pinned as the rules source. The project snapshot records Kaggle Environments 1.32.6 and a checksum-bound Linux x86-64 libcg.so. The paper archive does not redistribute that binary, replays, private trajectories, model or optimizer weights, Pokémon card images, or logos. Public strategy sources are cited rather than copied. Own-hand information is legitimate

TABLE II
RESULTS WITH EXPERIMENTAL UNIT AND DISPOSITION. “NOT MEASURED” IS A RESULT STATUS, NOT A BLANK TO BE INFERRED AS SUCCESS.

Status	Artifact / unit	Observation	Supported conclusion and disposition
Measured	Collision census / 18,412,973 option records	127,641 legacy groups; 255,398 affected records; 0 actionable repaired groups	Public-rule representation repair supported on the sealed corpus; no gameplay delta inferred
Receipt-verified	Full bootstrap / 25 epochs	57.88M decision visits; loss 1.7141; semantic accuracy 42.27%	Intended full-model workload completed; metrics are optimization diagnostics
Measured	H3 sidecar / 380,499 validation actions	94.18% H3 coverage; action MSE 0.0040745 vs. 0.0046184 baseline	Held-out prediction and support evidence; sidecar is not runtime policy
Historical	Kaggle submission 55487412	Public score 600.0; NO-RTP, search/MCTS off	Contextual platform observation; not a paired causal comparison
Measured	Iteration-2 formal audit / 900 games	76.83% skill-weighted; 74.39% lower bound; broader pipeline gate failed	Receipt-audited non-promotion evidence; continuous learner retained
Not measured	Iteration-3 formal holdout	900 games waived; 0 measured	Waiver only; no pass claim
Configured	H3 multi-arm evaluation	8,192 matched games per eligible arm required	Future protocol; no completed gameplay result at snapshot

for the acting seat, but opponent hidden information remains excluded; for this reason we use “acting-player-visible” rather than the overly narrow phrase “public only.”

Several limitations remain. First, zero repaired semantic collisions in this corpus do not prove that every future simulator option is covered. Schema versioning and fail-closed unknown handling remain necessary. Second, the literal card-743 eligibility rule makes the evidence Alakazam-centered. The causal encoder and factorized decoder may transfer, but the distribution of selection contexts, targets, and Prize plans may not. Third, a 600.0 public score entangles policy, deck, opponent pool, and platform variance. It is not an adapter ablation. Fourth, the iteration-2 holdout was numerically strong but failed a broader pipeline gate; the iteration-3 formal result is absent by explicit waiver. Fifth, the H3 sidecar has prediction and ESS/clipping evidence but no completed required multi-arm gameplay evaluation. Sixth, current checklist gates are inert, and recursive planning/RTP remains outside the packaged action path.

Finally, auditability carries cost. Exact 8,196-game iterations, full expert refresh, paired seat balance, immutable artifacts, fleet parity, and isolated formal evaluation slow experimentation and consume substantial compute and storage. We consider this cost justified when the alternative is an irreproducible strength claim whose legality, information boundary, or data lineage cannot be reconstructed.

IX. CONCLUSION

This work treats legal-action semantics, partial-observation safety, and evidence provenance as one controller contract. The simulator owns legality; the model ranks a complete factorized legal set from acting-player-visible information; a zero-bypass residual repairs supported public distinctions; and unsupported routes remain inert. The collision census demonstrates the value of that discipline, reducing 127,641 legacy groups to zero unresolved actionable repaired groups. Gameplay evidence remains intentionally bounded: the current record supports a reproducible direct-policy system and a verified representation repair, while causal strength gains await completed paired evaluation.

REFERENCES

- [1] L. P. Kaelbling, M. L. Littman, and A. R. Cassandra, “Planning and acting in partially observable stochastic domains,” *Artificial Intelligence*, vol. 101, no. 1–2, pp. 99–134, 1998.
- [2] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed. MIT Press, 2018.
- [3] Project Team, “Alakazam elmo rule-derivative contract, revision 27,” 2026, internal typed design authority; SHA-256 prefix 68c4e657.
- [4] J. Pineau, P. Vincent-Lamarre, K. Sinha, V. Lariviere, A. Beygelzimer, F. d’Alche Buc, E. Fox, and H. Larochelle, “Improving reproducibility in machine learning research,” *Journal of Machine Learning Research*, vol. 22, no. 164, pp. 1–20, 2021.
- [5] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, “Attention is all you need,” in *Advances in Neural Information Processing Systems*, vol. 30, 2017.
- [6] A. Y. Ng, D. Harada, and S. Russell, “Policy invariance under reward transformations: Theory and application to reward shaping,” in *Proceedings of the Sixteenth International Conference on Machine Learning*, 1999, pp. 278–287.
- [7] E. B. Wilson, “Probable inference, the law of succession, and statistical inference,” *Journal of the American Statistical Association*, vol. 22, no. 158, pp. 209–212, 1927.
- [8] TCGplayer, “Alakazam deck guide - pokémon tcg,” Online strategy guide, accessed 14 August 2026. [Online]. Available: <https://www.tcgplayer.com/content/article/Alakazam-Deck-Guide-Pok%C3%A9mon-TCG/7eb46b82-9dc5-40d8-adf9-28cca05f070f/>
- [9] PTCGO News, “Ptcgl alakazam deck guide,” Online strategy guide, accessed 14 August 2026. [Online]. Available: <https://ptcgonews.com/tips/ptcgl-alakazam-deck-guide/>
- [10] Kaggle, “Pokémon tcg ai battle: Simulation overview,” Competition overview, 2026, accessed 14 August 2026. [Online]. Available: <https://www.kaggle.com/competitions/pokemon-tcg-ai-battle/overview/description>